

NAVWEPS 00-80T-80
APPLICATION OF AERODYNAMICS
TO SPECIFIC PROBLEMS OF FLYING

usually means remaining over a particular spot on the ground, it shall be considered here as flight at zero airspeed. This is necessary because the aerodynamic characteristics of the rotor depend on its motion with respect to the air and not the ground. Hovering in a 20 knot wind is aerodynamically equivalent to flying at an airspeed of 20 knots in a no-wind condition, and the characteristics will be identical in the two conditions.

The first point to realize is that the rotor is subject to the same physical laws of aerodynamics and motion that govern flight of the fixed-wing airplane. The manner in which the rotor is subject to these laws is much more complicated due to the complex flow conditions.

Rotor lift can be explained by either of two methods. The first method, utilizing simple momentum theory based on Newton's Laws, merely states that lift results from the rotor accelerating a mass of air downward in the same way that the jet engine develops thrust by accelerating a mass of air out the tailpipe. The second method of viewing rotor lift concerns the pressure forces acting on the various sections of the blade from root to tip. The simple momentum theory is useful in determining only lift characteristics while the "blade element" theory gives drag as well as lift characteristics and is useful in giving a picture of the forces at work on the rotor. In the "blade element" theory, the blade is divided up into "blade elements" as shown in figure 6.15. The forces acting on each blade element are analyzed. Then the forces on all elements are summed up to give the characteristics of the whole rotor. The relative wind acting on each segment is the resultant of two velocity components: (1) the velocity due to the rotation of the blades about the hub and (2) the induced velocity, or downwash velocity caused by the rotor. The velocity due to rotation at a particular element is proportional to the rotor speed and the distance of the element from the rotor hub.

Thus, the velocity due to rotation varies linearly from zero at the hub to a maximum at the tip. A typical blade section with the forces acting on it is shown in figure 6.15.

A summation of the forces acting perpendicular to the plane of rotation (tip path plane) will determine the rotor thrust (or lift) characteristics while summation of the moments resulting from forces acting in the plane of rotation will determine the rotor torque characteristics. As a result of this analysis, the rotor thrust (or lift) is found to be proportional to the air density, a nondimensional thrust coefficient, and the square of the tip speed, or linear speed of the tip of the blade. The thrust coefficient is a function of the average blade section lift coefficient and the rotor solidity, which is the proportion of blade area to disc area. The lift coefficient is identical to that used in airplane aerodynamics while the solidity is analogous to the aspect ratio in airplane aerodynamics. The rotor torque is found to be proportional to a nondimensional torque coefficient, the air density, the disc area, the square of the tip speed, and the blade radius. The torque coefficient is dependent upon the average profile drag coefficient of the blades, the blade pitch angle, and the average lift coefficient of the blades. The torque can be thought to result from components of profile and induced drag forces acting on the blades, similar to those on an airplane.

As in the airplane, there is one angle of attack or blade pitch condition that will result in the most efficient operation. Unfortunately, the typical helicopter rotor operates at a near constant RPM and thus a constant true airspeed and cannot operate at this most efficient condition over a wide range of altitude and gross weight as the fixed-wing airplane. The airplane is able to maintain an efficient angle of attack at various altitudes and gross weights by flying at various airspeeds but the helicopter will operate with a near constant rotor velocity and vary blade angle to contend with variations in altitude and gross weight.